

Simultaneous Translation

Cz to En

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Introduction to the Problem

- Simultaneous translation generates words in the target language in real time.
- It interleaves **READ** (consumes source tokens) and **WRITE** (emits translations) actions.
- It does not wait for the full source sentence before starting translation.
- It trades-off translation quality and latency.

Other Approaches to the Problem

- **Wait-k** 🙅
 - Always wait until you have in total k more words on the input than you have sent to the output.
- **Hidden Markov transformer** 🥷
 - Use a transformer based model with a special loss to determine when we should wait - can use the same backbone as the translation model.
- **Ask an LLM** 🤖
 - Large language models should have a “working understanding” of translation and their ability for limited logical inference could be useful to determine optimal boundaries to wait for.

Chosen Approaches

- **Local Agreement**

- After the first output we keep a memory of the previous theory.
- Only output parts of the new theory that agree with the previous theory.
- Potentially when using beam search we could keep multiple theories but we did not try this.

- **AlignAtt**

- Looking at cross attention in a given layer/layers.
- We only output until we hit the output first token who's most attended to input token is in the f last inputs read.
- This hopefully causes us to wait until the model reads all the context it considers necessary to translate.

Evaluation Metrics

- **Bilingual Evaluation Understudy (BLEU)** 
 - Evaluates the quality of the translation compared to a golden text
- **Average Latency (AL)** 
 - Calculated the word delays for *input + prediction output* pair
 - Calculated the Average Latency given the word delays
- **Word Error Rate (WER) should be Translation Error Rate (TER)** 
 - Measures translation accuracy comparing predicted and golden text by the amount of substitutions, insertions, and deletions between the two

Finetuning and data preparation

- **Preparation of training dataset** 
 - From each parallel sentence pair we generate multiple prefix pairs
 - Start with the shorter sentence's first word
 - Add words from the longer sentence in proportion to their character-length
- **Example of the data** 

```
{"pref_source": "Jsme v", "pref_target": "We're fine,"}
```

```
{"pref_source": "Jsme v pořádku Margaret .", "pref_target":  
"We're fine, Margaret."}
```

```
{"pref_source": "Všechny oběti", "pref_target": "All the  
victims"}
```

```
{"pref_source": "Všechny", "pref_target": "All the"}
```

Finetuning and data preparation

- Finetuned Czech to English NMT model on sentence prefixes
- Finetuning data:
 - Training data: ~1.7M prefixes
 - Evaluation data: 5k prefixes
 - two random prefixes of one original sentence, train and eval contain prefixes from distinct original sentences
- Finetuned on NVIDIA H100 94GB (very cool 🧐)
- Before finetuning: **55.80 BLEU** 🤔
- After finetuning: **64.45 BLEU** 🚀
- The model is available online:
[🤗davidruda/opus-mt-cs-en-Prefix-Finetuned](https://huggingface.co/davidruda/opus-mt-cs-en-Prefix-Finetuned)

Results - Plain NMT

NMT Baseline

#Beams	1 (Wait At Begin)	2 (Wait At Begin)	3 (Wait At Begin)
1	BLEU: 11.43, AL: 7.75, WER: 0.87	BLEU: 13.09 , AL: 7.82, WER: 0.83	BLEU: 12.40, AL: 8.20, WER: 0.83
2	BLEU: 10.44, AL: 7.88, WER: 0.89	BLEU: 11.09, AL: 8.08, WER: 0.86	BLEU: 11.29, AL: 8.40, WER: 0.82
3	BLEU: 9.99, AL: 7.95, WER: 0.90	BLEU: 11.18, AL: 8.14, WER: 0.87	BLEU: 11.64, AL: 8.41, WER: 0.83

NMT Finetuned

#Beams	1 (Wait At Begin)	2 (Wait At Begin)	3 (Wait At Begin)
1	BLEU: 17.01, AL: 8.71, WER: 0.80	BLEU: 17.21, AL: 8.88, WER: 0.78	BLEU: 17.04, AL: 9.13, WER: 0.77
2	BLEU: 16.79, AL: 8.72, WER: 0.78	BLEU: 16.82, AL: 8.92, WER: 0.78	BLEU: 17.62 , AL: 9.06, WER: 0.76
3	BLEU: 16.84, AL: 8.70 , WER: 0.79	BLEU: 16.90, AL: 8.93, WER: 0.78	BLEU: 17.53, AL: 9.14, WER: 0.76

Results - Local Agreement

Local Agreement NMT Baseline

#Beams	1 (Wait At Begin)	2 (Wait At Begin)	3 (Wait At Begin)
1	BLEU: 11.43, AL: 7.75 , WER: 0.87	BLEU: 13.09 , AL: 7.82, WER: 0.83	BLEU: 12.40, AL: 8.20, WER: 0.83
2	BLEU: 10.44, AL: 7.88, WER: 0.89	BLEU: 11.09, AL: 8.08, WER: 0.86	BLEU: 11.29, AL: 8.40, WER: 0.82
3	BLEU: 9.99, AL: 7.95, WER: 0.90	BLEU: 11.18, AL: 8.14, WER: 0.87	BLEU: 11.64, AL: 8.41, WER: 0.83

Local Agreement NMT Finetuned

#Beams	1 (Wait At Begin)	2 (Wait At Begin)	3 (Wait At Begin)
1	BLEU: 15.14, AL: 9.40, WER: 0.77	BLEU: 15.40, AL: 9.44, WER: 0.76	BLEU: 16.13 , AL: 9.60, WER: 0.75
2	BLEU: 15.39, AL: 9.37, WER: 0.77	BLEU: 15.67, AL: 9.50, WER: 0.76	BLEU: 15.89, AL: 9.53, WER: 0.75
3	BLEU: 15.34, AL: 9.34 , WER: 0.77	BLEU: 15.15, AL: 9.47, WER: 0.76	BLEU: 16.07, AL: 9.53, WER: 0.74

Results - Local Agreement

Local Agreement EuroLLM

#Beams	1 (Wait At Begin)	2 (Wait At Begin)	3 (Wait At Begin)
1	BLEU: 11.33, AL: 9.33, WER: 0.79	BLEU: 11.62, AL: 9.73, WER: 0.79	BLEU: 11.02, AL: 10.25, WER: 0.79
2	BLEU: 11.34, AL: 9.29 , WER: 0.80	BLEU: 11.78 , AL: 9.75, WER: 0.79	BLEU: 11.44, AL: 10.23, WER: 0.80
3	BLEU: 11.14, AL: 9.33, WER: 0.80	BLEU: 11.58, AL: 9.73, WER: 0.79	BLEU: 11.55, AL: 10.22, WER: 0.79

Results - AlignAtt

AlignAtt NMT Baseline

Frame Size	4. (Layer)	3. (Layer)	2. (Layer)
1	BLEU: 11.53, AL: 8.90, WER: 0.78	BLEU: 11.19, AL: 8.70 , WER: 0.78	BLEU: 13.75, AL: 8.80, WER: 0.79
2	BLEU: 11.28, AL: 8.91, WER: 0.78	BLEU: 11.09, AL: 8.72, WER: 0.78	BLEU: 14.30, AL: 8.89, WER: 0.78
3	BLEU: 11.10, AL: 8.94, WER: 0.78	BLEU: 10.92, AL: 8.73, WER: 0.78	BLEU: 14.53 , AL: 9.06, WER: 0.77

AlignAtt NMT Finetuned

Frame Size	4. (Layer)	3. (Layer)	2. (Layer)
1	BLEU: 8.34, AL: 9.06, WER: 0.78	BLEU: 8.92, AL: 9.06, WER: 0.77	BLEU: 15.19, AL: 9.37, WER: 0.74
2	BLEU: 8.24, AL: 9.05 , WER: 0.78	BLEU: 8.80, AL: 9.08, WER: 0.77	BLEU: 15.27 , AL: 9.39, WER: 0.74
3	BLEU: 8.07, AL: 9.08, WER: 0.78	BLEU: 8.87, AL: 9.13, WER: 0.77	BLEU: 15.15, AL: 9.36, WER: 0.74

Results - Best per Table

	BLEU	WER	AL
NMT Finetuned	17.62	0.76	9.06
LA NMT Baseline	16.53	0.78	9.11
AligNatt NMT Finetuned	16.43	0.73	9.36
LA NMT Finetuned	16.07	0.74	9.53
AligNatt NMT Baseline	14.30	0.78	8.89
NMT Baseline	13.09	0.83	7.82
LA EuroLLM	11.33	0.79	9.33

Example Output - Golden Data

- Czech text

"V materiálu, který mám připravený, je krásný argument, že pokud omezíme pravomoc obecní policie právě v oblasti měření rychlosti nebo kontroly bezpečnosti silničního provozu a rychlosti, tak sice snížíme počet přestupků odhalených, na které se přišlo a které byly pokutovány, ale rozhodně nesnížíme počet přestupků spáchaných."

- English text

"In the material I have had prepared, there is a beautiful argument for limiting the powers of local police when it comes to gauging speed or observing the safety of road users, which will of course lower the number of discovered offenses, which would otherwise be found out about and fined, however it will not reduce the number of offenses themselves."

Example Output - Model Results

experiment name	final translation
LLM local agreement	In the material that I have prepared, there is a beautiful argument that if we limit the authority of the municipal police in just the area of measuring the speed or controlling the safety of road traffic and speed, then we will reduce
baseline alignatt	In the material I have prepared is beautiful argument that if we limit the authority of the municipal police in the area of speed measurement or control of road safety, then, although we will reduce the number of offences detected, to which we have been fined, but we are not going to reduce any
baseline local agreement	In the material I have prepared, there is a beautiful argument that if we limit the authority of the municipal police in the area of speed measurement, or road safety control, and speed, we will reduce the number of offences detected, to which we have come and which have been fined, but we certainly will not reduce any offences committed.
baseline wait beginning	In material I have prepared, it is beautiful argument, that if we limit the power of the municipal police just in the area of measurement of speed or control of safety of road traffic, and speed, we will be able to reduce the number of offences detected, to:
finetuned alignatt	In the material I have prepared, there is a beautiful argument that if we limit the authority of the municipal police in the area of measuring the speed or control of road traffic safety and speed, we will reduce the number of offences exposed to which were fined, but will not reduce those
finetuned local agreement	In the material I have prepared, there is a beautiful argument that if we limit the authority of the municipal police precisely in the area of speed measurement or road roadding road traffic and speed safety, sožlilitícítlící který předvětlící osvrátek.
finetuned wait beginning	In the material I have prepared, there is a beautiful argument that if we limit the authority of the municipal police in the area of speed measurement or safety control of road traffic and speed, so we will reduce the number of offences exposed to which they were fined, but we won't reduce those offences.

THE MOMENT YOU FINISH



YOUR PROJECT